

NISHANT SINGH

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PROFESSIONAL SUMMARY

Backend Engineer with 2 years of experience building Python backend platforms, **REST APIs, and infrastructure tools** for Cisco SD-WAN environments at Tata Communications — reducing manual effort by **65%** and cutting upgrade validation time from 2–3 hours to under 10 minutes. **Skilled** in FastAPI, Flask, PostgreSQL, Docker, and CI/CD, with deep networking knowledge enabling resilient backend solutions for complex infrastructure.

EXPERIENCE

Engineer – Cisco SD-WAN & Infrastructure Automation

Jul 2024 – Present

Tata Communications Limited, Chennai, India

- Developed Python backend services using FastAPI, Flask, and REST APIs for SD-WAN upgrade orchestration, infrastructure validation, and centralized log collection across 20+ operational workflows.
- Integrated **Cisco vManage REST APIs** for real-time device health checks, centralized infrastructure control, and automated operations across SD-WAN controllers and edge devices.
- Engineered automated **pre/post-upgrade validation pipelines** with 20+ health checks, reducing validation time from **2–3 hours to under 10 minutes** while improving deployment reliability and configuration consistency.
- Developed **web-based interfaces** with live execution tracking and real-time log streaming, reducing troubleshooting turnaround by **40%** and improving operational transparency.
- Automated repetitive operational tasks, cutting manual backup and troubleshooting effort by **65%** and improving operational response times.
- Developed backend applications using Python and PostgreSQL, optimizing database queries for performance and maintainability.
- Operated in Linux environments using Git/GitHub for version control, collaborative development, and CI/CD pipeline integration.
- Leveraged networking expertise — TCP/IP, DNS, HTTP/HTTPS, DHCP, NAT, routing, and SD-WAN architecture — to inform automation design and system-level troubleshooting.

PROJECTS

Cisco SD-WAN Change Management Platform

- Designed a **web-based platform** using Flask, Playwright, and APIs; reduced manual effort by **60%** and improved deployment efficiency across validation environments.
- Implemented automated upgrade execution, version management, and **20+ pre/post validation checks**, reducing upgrade validation time from 2–3 hours to under 10 minutes.
- Built web-based interfaces for automation execution, workflow visibility, and live operational log streaming.

Multi-Hop Network Configuration Backup & Drift Detection

- Developed Python-based automation using **Paramiko** to automate CLI login, configuration backup, and infrastructure validation across SD-WAN devices and controllers.
- Implemented centralized log collection and version-controlled backup workflows using **Git/GitHub**, reducing backup time to under 5 minutes and improving recovery reliability.

TECHNICAL SKILLS

Programming Languages

Python

Programming Concepts

Object-Oriented Programming (OOP), Async Programming

Backend Development & APIs

FastAPI, Flask, REST APIs, JSON

Automation & Testing

Playwright, Paramiko

Databases

PostgreSQL, MySQL

Cloud, DevOps & Containers

Docker, Linux, Git, GitHub, GitHub Actions, CI/CD

Security & API Management

JWT Authentication (Learning), API Security

Networking & Infrastructure

Cisco SD-WAN, TCP/IP, DNS, HTTP/HTTPS

CERTIFICATIONS

Programming with Python Professional Certificate – OpenEDG	July 2026
Career Essentials in GitHub Professional Certificate – GitHub	June 2026
Ubuntu Linux Professional Certificate – Canonical	June 2026
Docker Foundations Professional certificate – Docker	June 2026
Catalyst SD-WAN Foundations – LinkedIn Learning	Oct 2024
Introduction to SD-WAN – LinkedIn Learning	Oct 2024
Meta Front-End Developer Professional Certificate – Meta	Jan 2024

EDUCATION

Bachelor of Technology in Computer Science and Engineering	2020 – 2024
S.R.M. Institute of Science and Technology CGPA: 8.88 / 10	